

Problem Description

A methanol vapor stream at 12,750 kg/hr is coming out at 95 °C from a distillation column.

It is required to design a countercurrent shell and tube heat exchanger in order to condense this methanol stream upto 40 °C in liquid state. Cold water at 25 °C with a flow rate of 70 kg/s is available as the coolant.

- a. Use shortcut method to estimate the basic Heat Exchanger parameters.
- b. Use detailed methods to design the geometry of the heat exchanger.
- c. Propose a more suitable heat exchanger design for this application.